

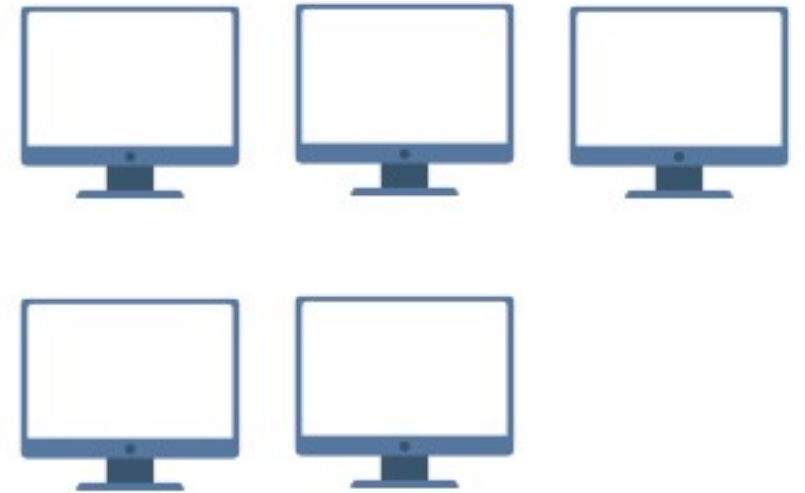
1. **Decomposing a Cloud:**
Virtualization, Containerization and Serverless Execution
2. **Beyond Basic Storage: Advanced Azure Services/features**

The cloud is changing the way applications are designed:

- Instead of monoliths, applications are decomposed into smaller, decentralized services.
- These services communicate through APIs or by using asynchronous messaging or eventing.
- Applications **scale horizontally**, adding new instances as demand requires.

Horizontal Scaling

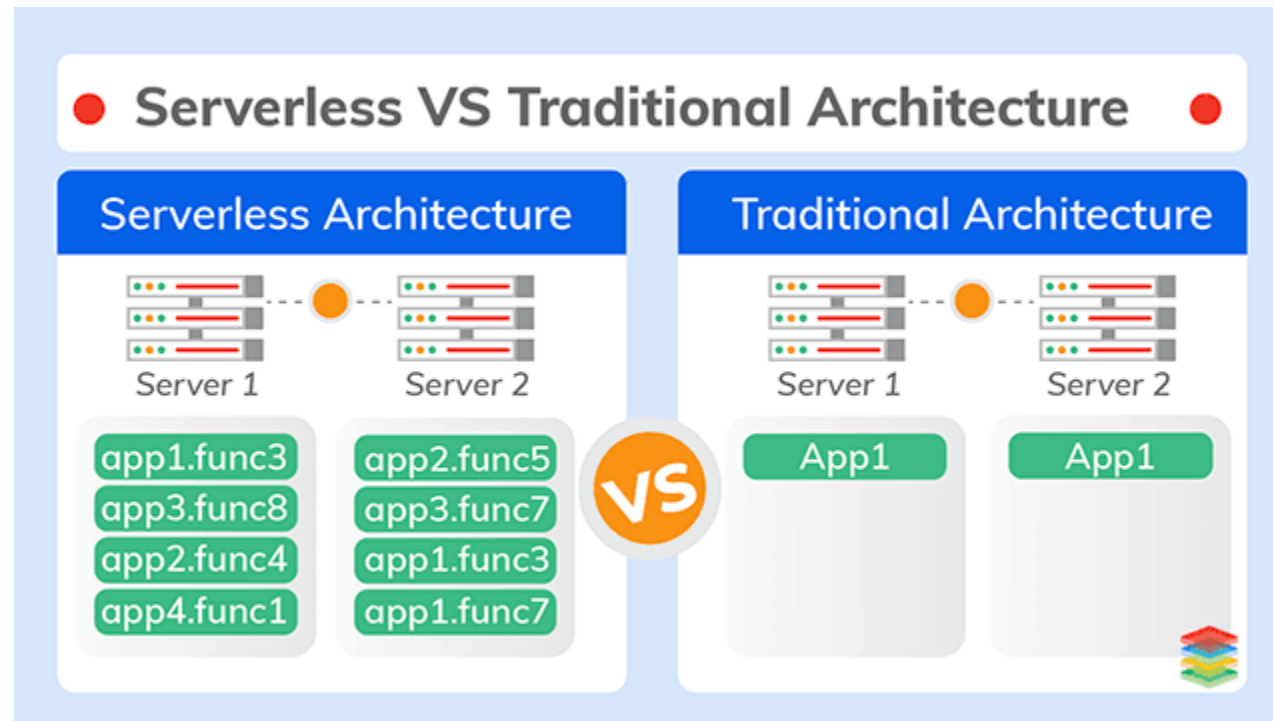
Add more instances



Essentially, we have a lot of data to store and process in order to keep hyperscale services running. Previous approaches cannot scale anymore.



Functions/Serverless Development



Functions/Serverless

Azure serverless computing includes the **Azure Functions, Logic Apps, Container Instances, & App Service**.

With serverless computing, your cloud platform takes care of managing the server infrastructure and the allocation and deallocation of resources based on demand (scalability).

- Infrastructure isn't your responsibility.
- Scaling and performance are handled automatically.
- You're billed only for the exact resources you use.
- There's no need to even reserve capacity.

Service	Primary Use	When to Choose
Azure Functions	Code execution	Need to run custom code responses
Logic Apps	Workflow automation	Integrating multiple services visually
Container Instances	Container runtime	Running existing containerized apps
App Service	Web applications	Traditional web apps with serverless scaling



Functions/Serverless

Functions are ideal when you're concerned only about the code running your service and not the underlying platform or infrastructure.

They're commonly used **when you need to perform work in response to an event** (often via a REST request), **timer**, or **message from another service**, and when that work can be completed quickly, within seconds or less.



Event-driven scale

Serverless computing is an excellent fit for workloads that respond to incoming events. Events include triggers by:

- Timers, for example, if a function needs to run every day at 10:00 AM UTC.
- HTTP loads, for example, API and webhook scenarios.
- Queues, for example, with order processing.



Micro-billing

Traditional computing bills for a block of time like paying a monthly or annual rate for website hosting.

This method of billing is convenient but isn't always cost effective. Even if a customer's website gets only **one hit a day**, they **still pay for a full day's worth of availability**.

With serverless computing, they **pay only for the time their code runs**. If no active function executions occur, they're not charged. For example, if the code runs once a day for two minutes, they're charged for one execution and two minutes of computing time.



Functions/Serverless

Azure has two main implementations of serverless compute.

Azure Functions

Functions can execute code in almost any modern language.



Azure Logic Apps

Logic apps are designed in a web-based designer and can execute logic triggered by Azure services without writing any code.



Azure Functions

When you're concerned only about the code running your service, and not the underlying platform or infrastructure, using Azure Functions is ideal.

Functions are commonly used when you need to perform work in response to an event (often via a REST request), timer, or message from another Azure service, and when that work can be completed quickly, within seconds or less.

Functions scale automatically based on demand, so they're a compelling choice when demand is variable.

For example, you might receive messages from an IoT solution that's used to monitor a fleet of delivery vehicles. You'll likely have more data arriving during business hours.



Azure Logic Apps

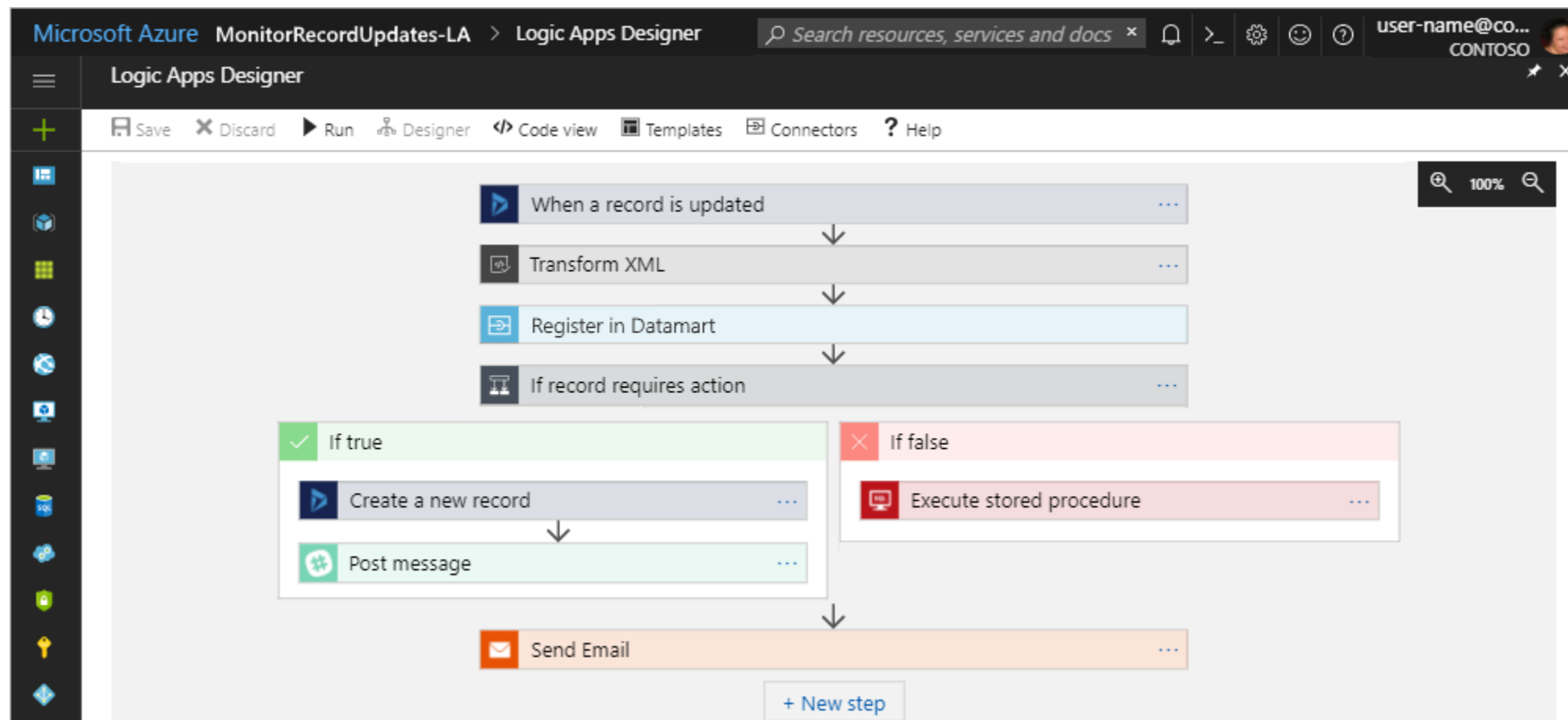
Logic apps are similar to functions. Both enable you to trigger logic based on an event. Where functions execute code, **logic apps execute workflows that are designed to automate business scenarios** and are built from predefined logic blocks.

Every Azure logic app **workflow starts with a trigger**, which fires when a specific event happens or when newly available data meets specific criteria.

Many triggers include basic scheduling capabilities, so developers can specify how regularly their workloads will run. Each time the trigger fires, the Logic Apps engine creates a logic app instance that runs the actions in the workflow. These actions can also include data conversions and flow controls, such as conditional statements, switch statements, loops, and branching.

You **create logic app workflows by using a visual designer** on the Azure portal or in Visual Studio. The workflows are persisted as a JSON file with a known workflow schema.





Functions vs. Logic Apps

	Functions	Logic Apps
State	Normally stateless, but Durable Functions provide state	Stateful
Development	Code-first (imperative)	Designer-first (declarative)
Connectivity	About a dozen built-in binding types. Write code for custom bindings	Large collection of connectors. Enterprise Integration Pack for B2B scenarios. Build custom connectors
Actions	Each activity is an Azure function. Write code for activity functions	Large collection of ready-made actions
Monitoring	Azure Application Insights	Azure portal, Log Analytics
Management	REST API, Visual Studio	Azure portal, REST API, PowerShell, Visual Studio

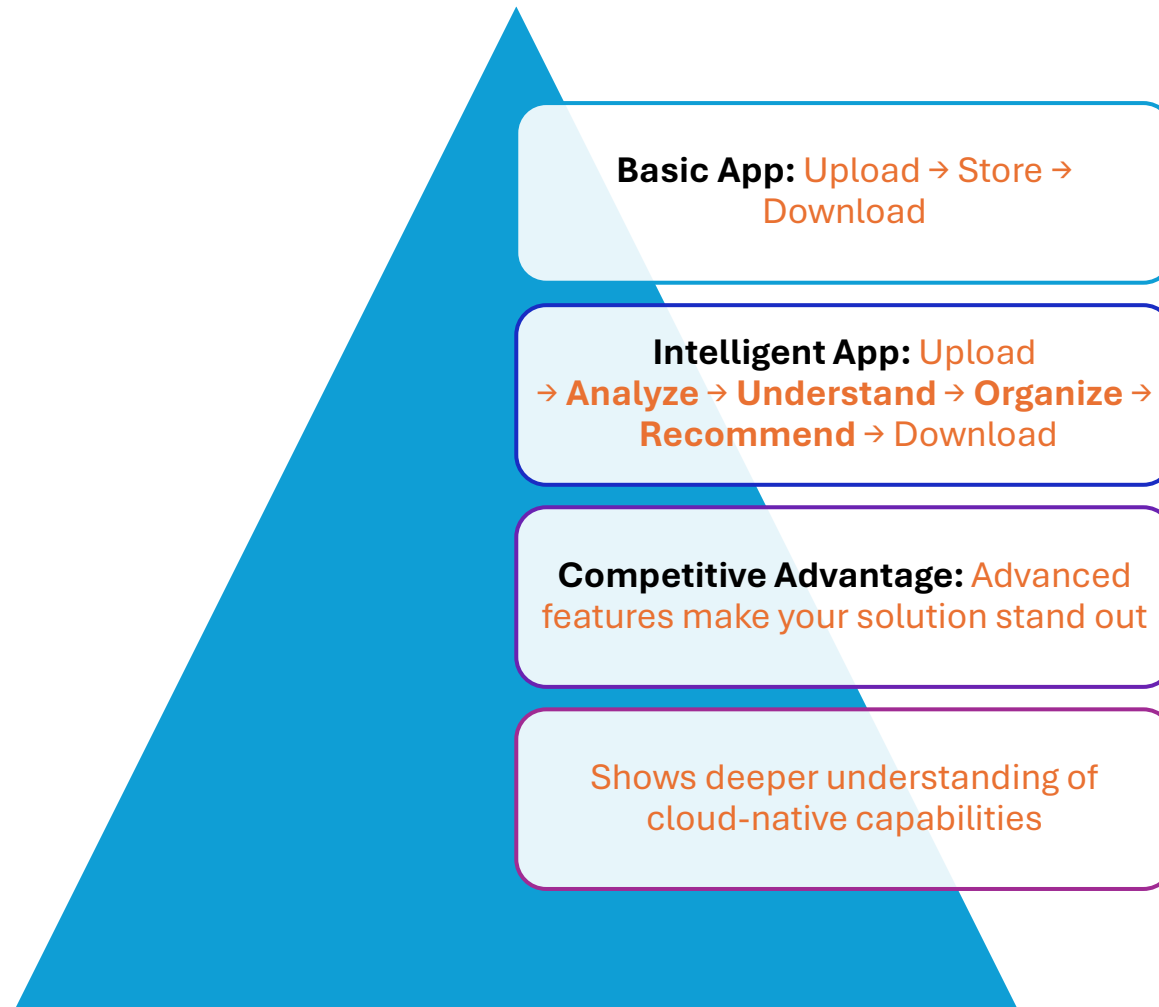
Beyond Basic Storage: Advanced Azure Services

Adding AI, Analytics & Intelligence to Your Multimedia Platform

Key Question: How can you make your app smarter than just upload/download (CRUD) ?

CW1 would require you to **just** list, ideally 2 of, the advance features, which are related to your use case.

Why Advanced Services?



[Home](#) >

Free services ...

Services that include monthly free amounts for 12 months

For 12 months after signup, new customers can use up to the specified monthly free amount of each of these services without getting charged. **Service usage is billed at the pay-as-you-go rate** [learn more, see the Azure free account FAQ](#)



Windows Virtual Machine
COMPUTE

750 hours

each of B1s and B2ats v2 (AMD-based) burstable VMs.

Create Windows virtual machines (VMs) in seconds to meet your workload and budget needs. [Learn more](#)

Create



Linux Virtual Machine
COMPUTE

750 hours

each of B1s and B2ats v2 (AMD-based) burstable VMs.

Create Linux virtual machines (VMs) in seconds to meet your workload and budget needs. [Learn more](#)

Create



Azure Managed Disks
STORAGE

64 GB x 2

(P6) solid state drives SSD storage, plus 1 GB snapshot and 2 million I/O operations

Get high performance, durable block storage for Azure Virtual Machines with simplified management. [Learn more](#)



Azure Blob Storage
STORAGE

5 GB

locally redundant storage (LRS) hot block with 20,000 read and 10,000 write operations

Use massively-scalable object storage for any type of unstructured data. [Learn more](#)

Create



Azure Files
STORAGE

100GB

of LRS transaction optimized, hot, and cool files. 2 million read, list, and other file operations

Migrate to simple, distributed, cross-platform file storage without changing



Key Vault
SECURITY

10,000 transactions

RSA 2048-bit keys or secret operations, Standard tier.

Safeguard and maintain control of keys and other secrets. [Learn more](#)



Azure Media Services Encodin...
MEDIA

20 output minutes

Standard encoder video or audio source file encoding.

Index, package, protect, and stream video and audio at scale. [Learn more](#)



Azure Database for MySQL
DATABASES

750 hours

of Flexible Server—Burstable B1MS Instance, 32 GB storage, and 32 GB backup storage

Host a fully managed, scalable MySQL database in Azure. [Learn more](#)

Free services ...



AI Custom Vision
AI + MACHINE LEARNING

10,000

predictions S0 tier, one training hour, two projects with 5,000 training images each

Easily customize computer vision models for your unique use case. [Learn more](#)

Create



Face
AI + MACHINE LEARNING

30,000

transactions S0 tier

Detect and identify people and emotions in images. [Learn more](#)

Create



AI Document Intelligence
AI + MACHINE LEARNING

500

pages S0 tier

Automate the extraction of text, key/value pairs, and tables from your documents. [Learn more](#)



Language Understanding
AI + MACHINE LEARNING

10,000

text request transactions S0 tier

Build natural language understanding into apps, bots, and IoT devices. [Learn more](#)

Create



AI Personalizer
AI + MACHINE LEARNING

50,000

transactions S0 tier

Deliver rich, personalized experiences for every user. [Learn more](#)

Create



Service Bus
DATABASES

750 hours, 13 million operations

Standard tier base unit

Get reliable cloud messaging as a service with simple hybrid integration. [Learn more](#)

Create



Container Registry
CONTAINERS

1 Standard tier registry

With 100 GB storage and 10 webhooks

Store and manage container images across all types of Azure deployments. [Learn more](#)

Create



Azure Load Balancer
NETWORKING

750 hours, 15 GB

of data processing, and up to five rules with Standard Load Balancer

Add scale, availability, and network performance to your applications instantly. [Learn more](#)

Create

[Home](#) >

Free services



Application Insights
DEVELOPER TOOLS

Unlimited

nodes

Monitor the use and performance of live apps running on an unlimited number of hosts or devices. [Learn more](#)

Create



DevTest Labs
DEVELOPER TOOLS

Free

Enable fast, easy, and lean dev-test environments [Learn more](#)

Create



Security Center
SECURITY + IDENTITY

Free

Prevent, detect, and respond to threats with increased visibility and control over the security of your Azure resources. [Learn more](#)



Advisor
MONITORING + MANAGEMENT

Unlimited

Get personalized recommendations to Azure best practices. [Learn more](#)



Microsoft IoT Hub
INTERNET OF THINGS

8,000

messages per day and .5 KB message meter size of Free edition

Connect, monitor, and manage billions of IoT assets. [Learn more](#)

Create



Load Balancer
NETWORKING

Free

public load balanced IP(VIP)

Add scale, availability, and network performance to your applications instantly. [Learn more](#)

Create



Data Factory
DATABASES

5

activities low frequency

Compose and manage data services at scale. [Learn more](#)

Create



AI Search
AI + MACHINE LEARNING

50 MB

storage for 10,000 hosted documents and 3 indexes per service

Include a cloud search service in your web and mobile application. [Learn more](#)

Create



Batch
COMPUTE

Free

Scale your application in the cloud with job orchestration and scheduling for HPC applications. [Learn more](#)

Create



Notification Hubs
MOBILE

1,000,000

push notifications with free namespace

Send push notifications to any platform from any back end. [Learn more](#)

Create



Automation
MONITORING + MANAGEMENT

500 minutes

of job run time

Simplify cloud management with process automation. [Learn more](#)

Create



Data Catalog
DATA + ANALYTICS

Unlimited

users

Get more value from your enterprise data assets. [Learn more](#)

Create



Face
AI + MACHINE LEARNING

30,000

transactions of Free instance



Virtual Network
NETWORKING

50

virtual networks



Inter-VNET Data Transfer
NETWORKING

Inbound only



Bandwidth (Data Transfer)
NETWORKING

100 GB

outbound



Machine Learning
AI + MACHINE LEARNING

Free



Azure Logic Apps
INTEGRATION

4,000

built-in actions with the Consumption plan

Category1: AI & Cognitive Services

Service	What It Does	Use Case for Multimedia Platform
Computer Vision	Analyzes image content	Auto-tag images, detect inappropriate content
Face API	Detects and recognizes faces	Blur faces for privacy, count people in photos
Video Indexer	Analyzes video content	Extract scenes, detect celebrities, generate thumbnails
Speech Service	Converts speech to text	Transcribe audio/video files, generate subtitles, Q&A maker
Text Analytics	Analyzes text sentiment	Analyse user comment, auto-moderate descriptions
Content Moderator	Detects offensive content	Auto-flag inappropriate uploads

Category2: Analytics & Dashboards

Service	What It Does	Use Case for Multimedia Platform
Azure Metrics	Built-in monitoring	Track uploads, storage usage, user activity
Azure Dashboards	Custom visualization	Show popular content, user engagement metrics
Application Insights	Performance monitoring	Monitor API response times, error rates
Power BI Embedded	Advanced analytics	Business intelligence on user behavior patterns

Category3: Advanced Storage & CDN

Service	What It Does	Use Case for Multimedia Platform
Azure CDN	Global content delivery	Faster media streaming worldwide
Azure Media Services	Professional video processing	Adaptive bitrate streaming, encoding
Azure Search	Intelligent search	Search inside images/videos using AI metadata
Service	What It Does	Use Case for Multimedia Platform

Overview

What is Azure Cognitive Services?

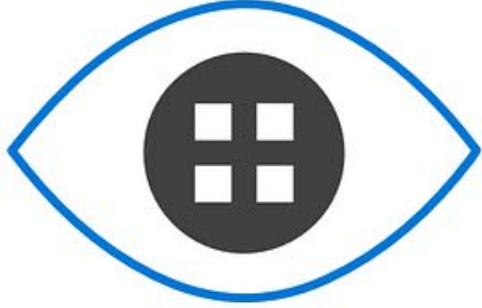
- **All it takes is an API call to embed the ability to see, hear, speak, search, understand, and accelerate decision-making into your apps.** Enable developers of all skill levels to easily add AI capabilities to their apps.



Overview

What is Azure Cognitive Services?

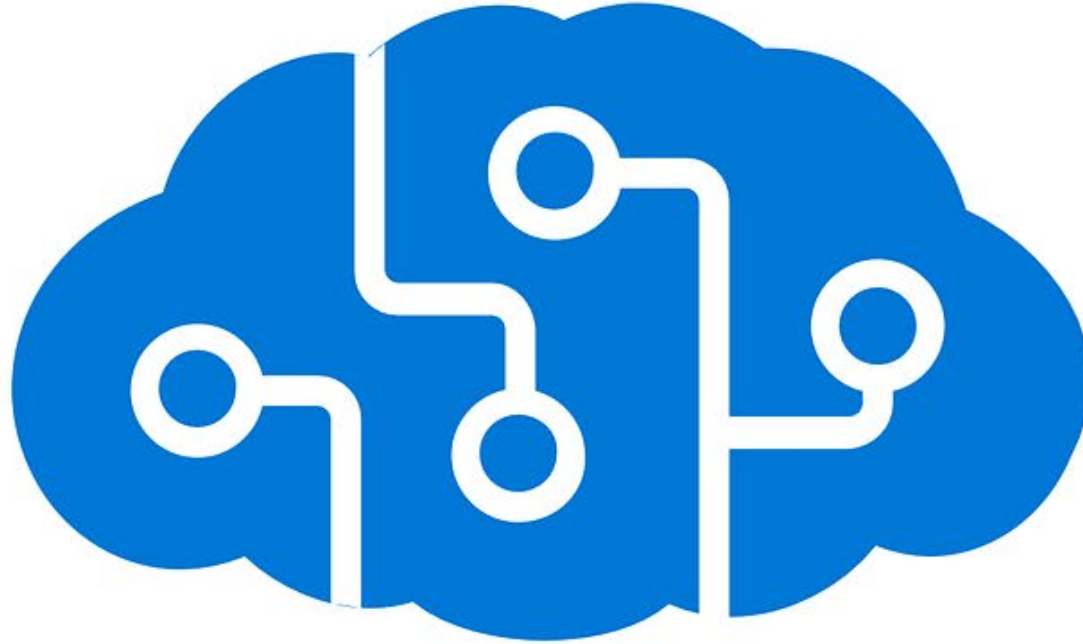
- **All it takes is an API call to embed the ability to see, hear, speak, search, understand, and accelerate decision-making into your apps.** Enable developers of all skill levels to easily add AI capabilities to their apps.



Vision



Language



Microsoft Azure Cognitive Services



Speech



Knowledge



Vision- Azure Cognitive Services

Azure Cognitive Services

Vision –*Computer Vision*

The cloud-based Computer Vision API provides developers with access to advanced algorithms for processing images and returning information.

By uploading an image or specifying an image URL, Microsoft Computer Vision algorithms can analyse visual content in different ways based on inputs and user choices.



Text extraction (OCR)

Extract printed and handwritten text from images and documents with mixed languages and writing styles.



Image understanding

Pull from a rich ontology of more than 10,000 concepts and objects to generate value from your visual assets.



Spatial analysis

Analyze how people move in a space in real time for occupancy count, social distancing and face mask detection.



Flexible deployment

Run Computer Vision in the cloud or on the edge, in containers.

Azure Cognitive Services

Vision –Computer Vision

Transform your processes

Apply these Computer Vision features to streamline processes, such as robotic process automation and digital asset management.

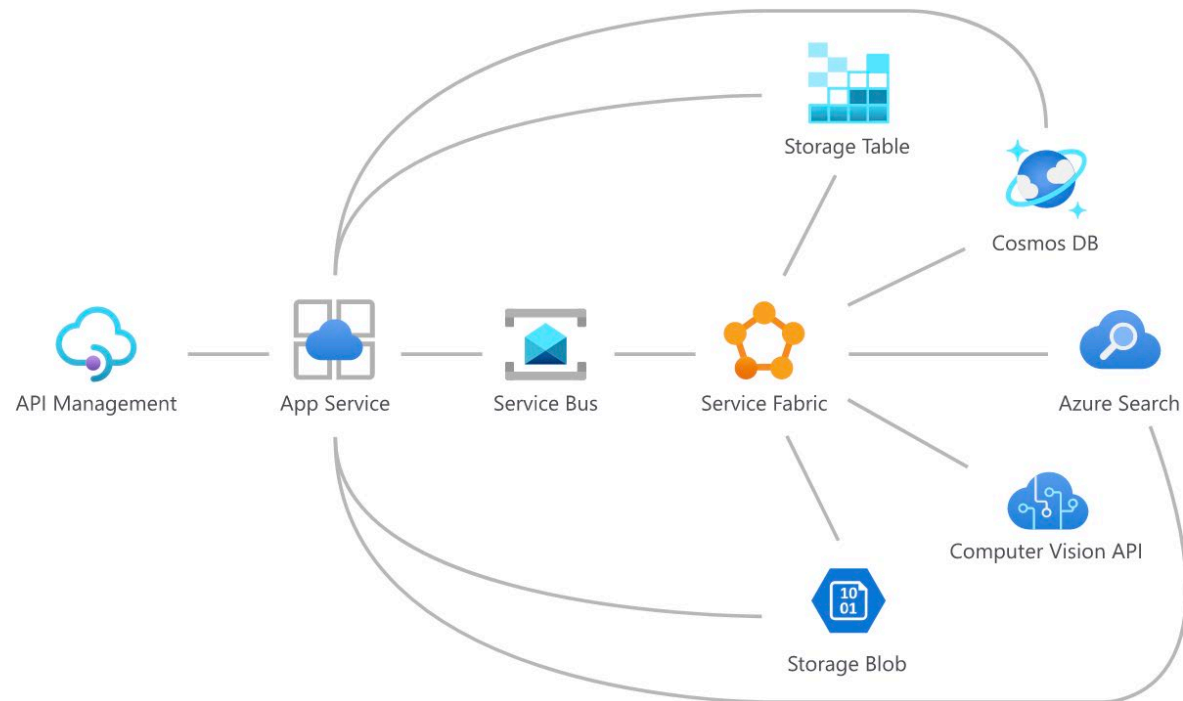


Image Classification

Overview –How images are stored



0	2	15	0	0	11	10	0	0	0	0	9	9	0	0	0
0	0	0	4	60	157	236	255	255	177	95	61	32	0	0	29
0	10	16	115	238	255	244	245	243	250	249	255	222	103	10	0
0	14	170	255	255	244	254	255	253	245	255	249	253	251	124	1
2	98	255	228	255	251	254	211	141	116	122	215	251	238	255	49
13	217	243	255	155	33	226	52	2	0	10	13	232	255	255	36
16	229	252	254	49	12	0	0	7	7	0	70	237	252	235	62
6	141	245	255	212	25	11	9	3	0	115	236	243	255	137	0
0	87	252	250	248	215	60	0	1	121	252	255	248	144	6	0
0	13	113	255	255	245	255	182	181	248	252	242	208	36	0	19
1	0	5	117	251	255	241	255	247	255	241	162	17	0	7	0
0	0	0	4	58	251	255	246	254	253	255	120	11	0	1	0
0	0	4	97	255	255	255	248	252	255	244	255	182	10	0	4
0	22	206	252	246	251	241	100	24	113	255	245	255	194	9	0
0	111	255	242	255	158	24	0	0	6	39	255	232	230	56	0
0	218	251	250	137	7	11	0	0	0	2	62	255	250	125	3
0	173	255	255	101	9	20	0	13	3	13	182	251	245	61	0
0	107	251	241	255	230	98	55	19	118	217	248	253	255	52	4
0	18	146	250	255	247	255	255	255	249	255	240	255	129	0	5
0	0	23	113	215	255	250	248	255	255	248	248	118	14	12	0
0	0	6	1	0	52	153	233	255	252	147	37	0	0	4	1
0	0	5	5	0	0	0	0	0	14	1	0	6	6	0	0

```
0 2 15 0 0 11 10 0 0 0 0 9 9 0 0 0
0 0 0 4 60 157 236 255 255 177 95 61 32 0 0 29
0 10 16 115 238 255 244 245 243 250 249 255 222 103 10 0
0 14 170 255 255 244 254 255 253 245 255 249 253 251 124 1
2 98 255 228 255 251 254 211 141 116 122 215 251 238 255 49
13 217 243 255 155 33 226 52 2 0 10 13 232 255 255 36
16 229 252 254 49 12 0 0 7 7 0 70 237 252 235 62
6 141 245 255 212 25 11 9 3 0 115 236 243 255 137 0
0 87 252 250 248 215 60 0 1 121 252 255 248 144 6 0
0 13 113 255 255 245 255 182 181 248 252 242 208 36 0 19
1 0 5 117 251 255 241 255 247 255 241 162 17 0 7 0
0 0 0 4 58 251 255 246 254 253 255 120 11 0 1 0
0 0 4 97 255 255 255 248 252 255 244 255 182 10 0 4
0 22 206 252 246 251 241 100 24 113 255 245 255 194 9 0
0 111 255 242 255 158 24 0 0 6 39 255 232 230 56 0
0 218 251 250 137 7 11 0 0 0 2 62 255 250 125 3
0 173 255 255 101 9 20 0 13 3 13 182 251 245 61 0
0 107 251 241 255 230 98 55 19 118 217 248 253 255 52 4
0 18 146 250 255 247 255 255 255 249 255 240 255 129 0 5
0 0 23 113 215 255 250 248 255 255 248 248 118 14 12 0
0 0 6 1 0 52 153 233 255 252 147 37 0 0 4 1
0 0 5 5 0 0 0 0 0 14 1 0 6 6 0 0
```

Image Classification

Overview –How images are stored



Colour Image



Image Classification

Overview – Feature extraction

You must have recognized the objects in an instant – a dog, a car and a cat. What are the features that you considered while differentiating each of these images? **The shape could be one important factor, followed by colour, or size.**

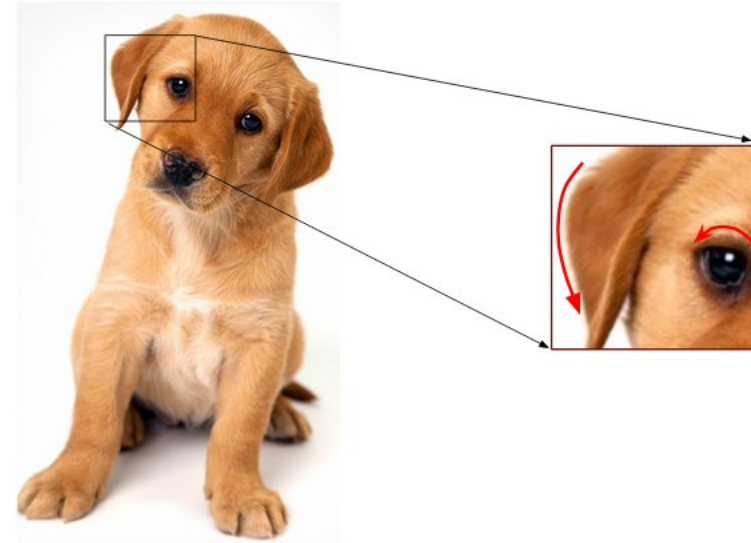
What if the machine could also identify the shape as we do?



Image Classification

Overview – Feature extraction

Edge is basically where there is a sharp change in colour. Look at the below image:



We could identify the edge because there was a change in colour from white to brown (in the right image) and brown to black (in the left). And as we know, an image is represented in the form of numbers. So, we will look for pixels around which there is a drastic change in the pixel values.

Image Classification

Overview – Feature extraction

- Let's say we have the following matrix for the image:

121	10	78	96	125
48	152	68	125	111
145	78	85	89	65
154	214	56	200	66
214	87	45	102	45

To identify if a pixel is an edge or not, we will simply subtract the values on either side of the pixel. For this example, we have the highlighted value of 85. We will find the difference between the values 89 and 78. Since this difference is not very large, we can say that there is no edge around this pixel.

- Now consider the pixel 125 highlighted in the below image:

121	10	78	96	125
48	152	68	125	111
145	78	85	89	65
154	214	56	200	66
214	87	45	102	45

Image Classification

Overview –Feature extraction

- Since the difference between the values on either side of this pixel is large, we can conclude that there is a significant transition at this pixel and hence it is an edge. Now the question is, do we have to do this step manually?
- No! There are various kernels that can be used to highlight the edges in an image. The method we just discussed can also be achieved using the Prewitt kernel (in the x-direction). Given below is the Prewitt kernel:

-1	0	1
-1	0	1
-1	0	1

Prewitt Kernel
X Direction

-1	-1	-1
0	0	0
1	1	1

Prewitt Kernel
Y Direction

-1	0	1
-2	0	2
-1	0	1

Sobel Kernel
X Direction

-1	-2	-1
0	0	0
1	2	1

Sobel Kernel
Y Direction

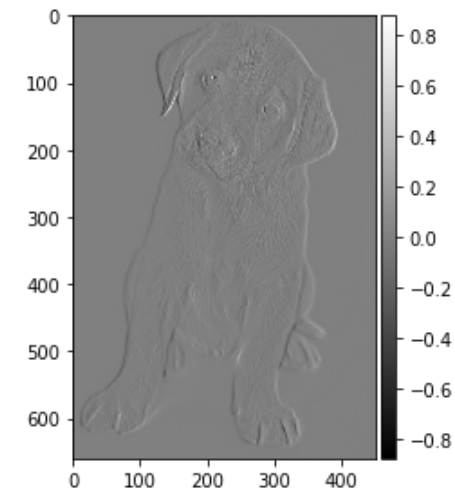
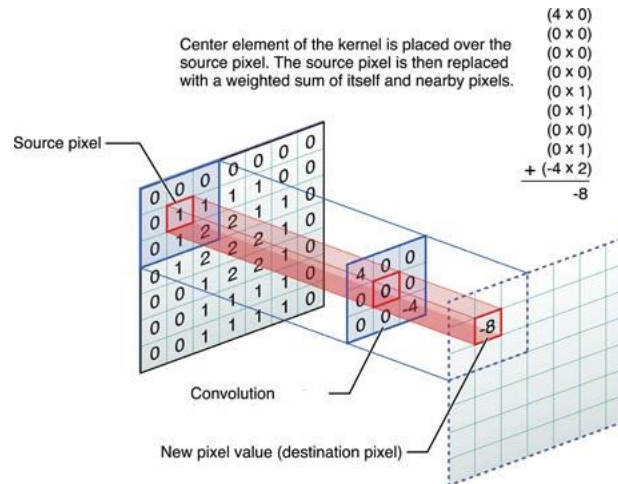
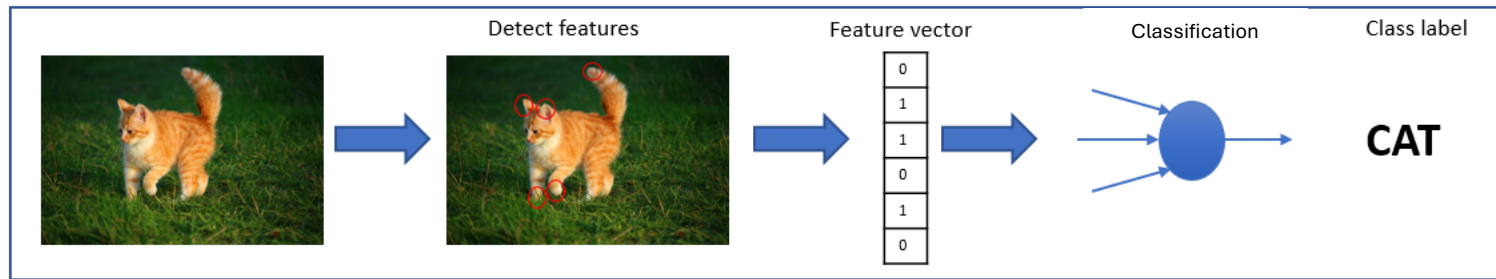


Image Classification

Overview – Feature extraction

Traditional Computer Vision



Deep learning

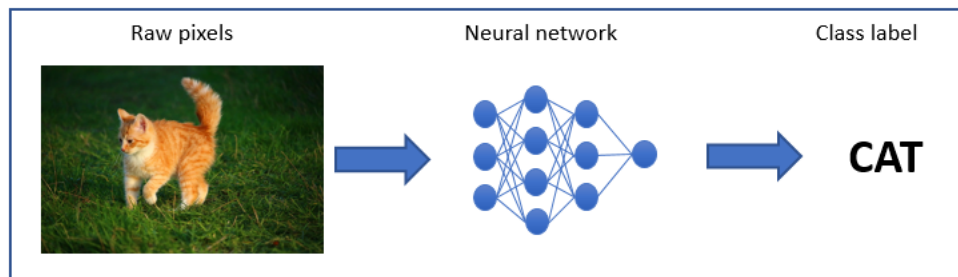
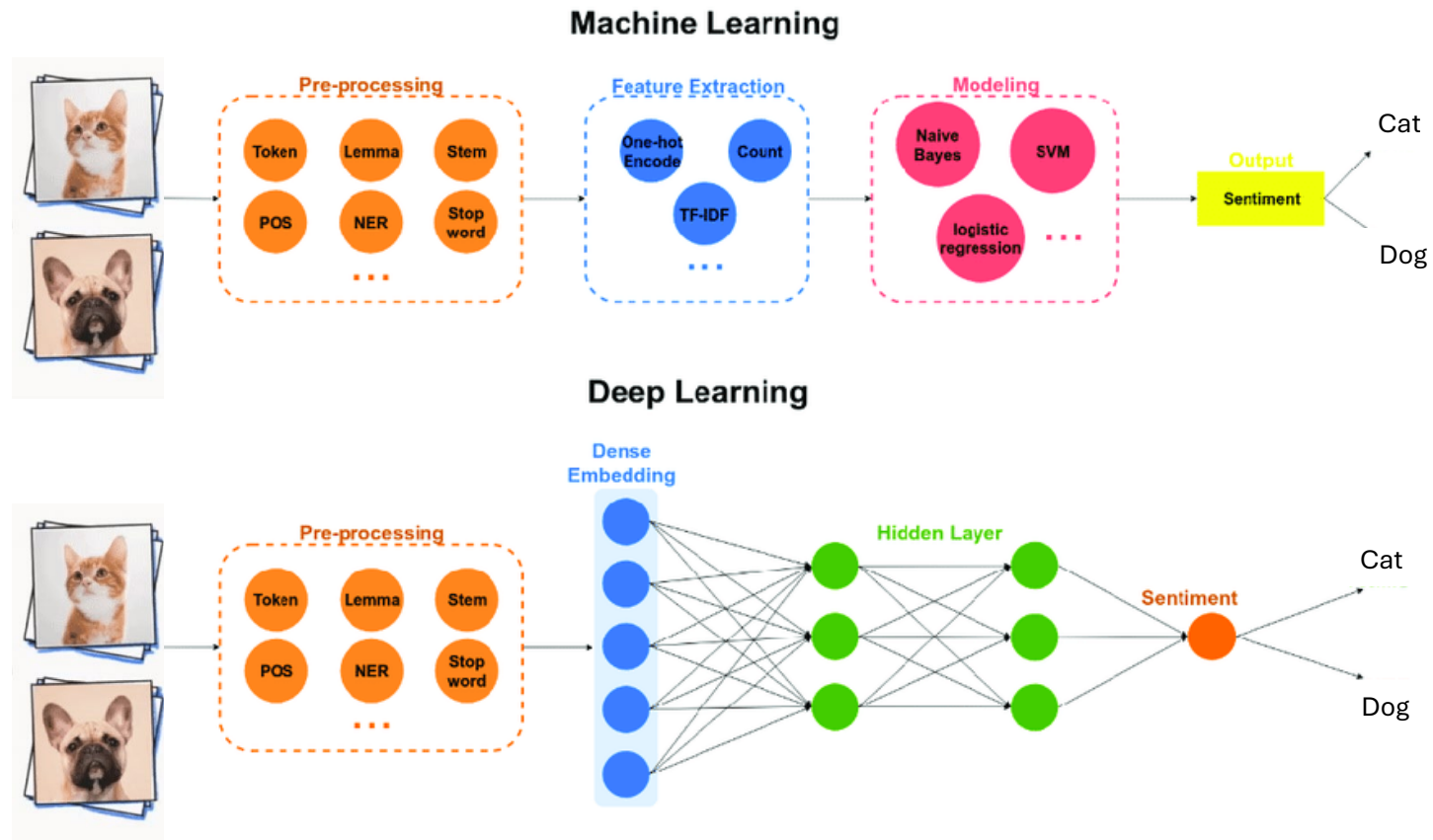


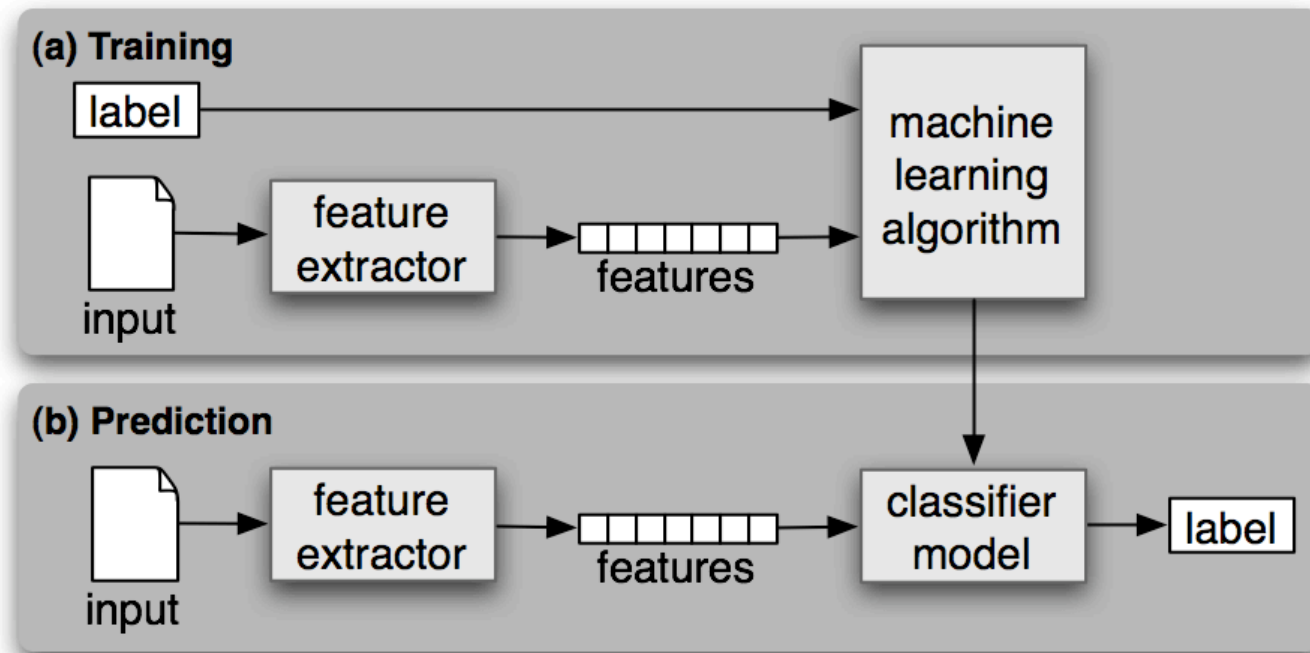
Image Classification

Overview – Feature extraction



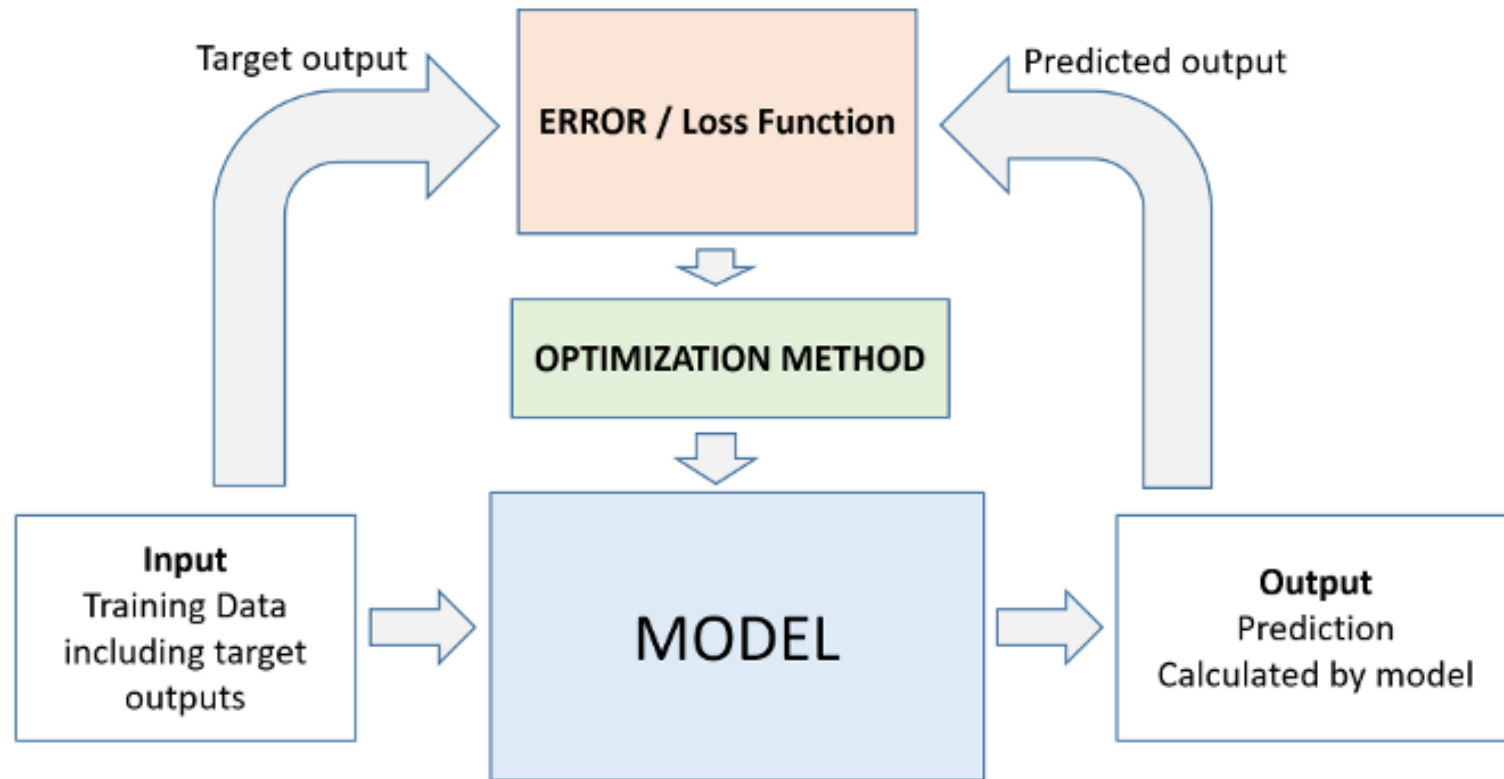
Machine Learning

Training and Prediction



Machine Learning

Training and Prediction



Azure Cognitive Services

Vision –Face API

Embed facial recognition into your apps for a seamless and highly secured user experience. Detect one or more human faces along with attributes such as pose, face coverings, and face location, including 27 landmarks for each face in the image.



Detect the location of a face in an image

Locate, crop, or blur faces to protect privacy using the number of faces and face rectangle coordinates. [Try it on the API console.](#)



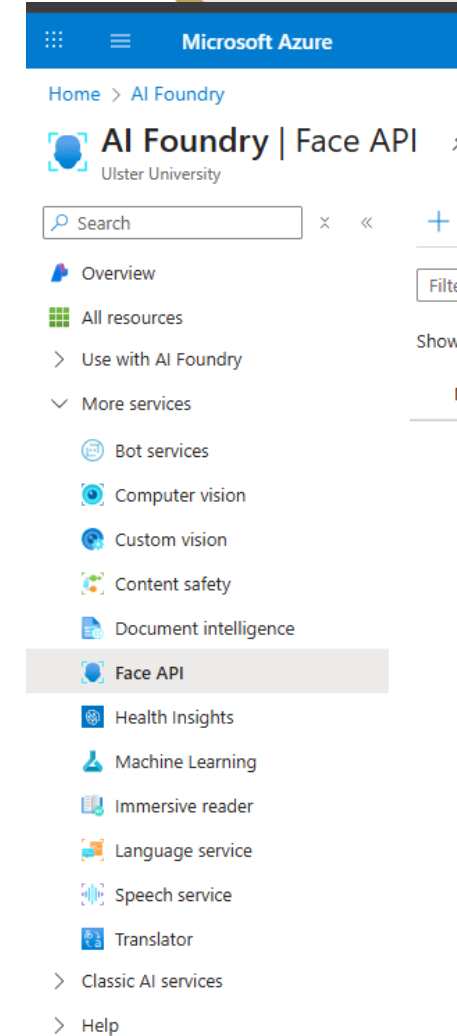
Detect 27 landmarks for each face, including eye position

Observe potential intent using precise labels, coordinates, and 3D head pose. [Learn more about face detection and attribute data.](#)



Analyze faces for attributes, including mask detection

Determine whether someone is wearing a face mask, and whether the mask covers both the nose and mouth, with the face mask attribute. [Try it on the API console.](#)





Decision

Azure Cognitive Services

Home > AI Foundry

AI Foundry | Anomaly detector (classic)

Ulster University

Discover key trends in resource metrics

Generate Terraform templates for selected resources

What platform metrics

Search

«

+ Create

Manage deleted resources

Manage view

Refresh

Export to CSV

Open query

Assign tags

Delete

- Overview
- All resources
- Use with AI Foundry
- Azure AI Foundry
- AI Hubs
- Azure OpenAI
- AI Search
- More services
- Bot services
- Computer vision
- Custom vision
- Content safety
- Document intelligence
- Face API
- Health Insights

Filter for any field...

Subscription equals all

Type equals all

Resource group equals all

Location equals all

Add filter

Showing 0 to 0 of 0 records.

Name ↑↓	Kind ↑↓	Location ↑↓	Custom Domain Name
---------	---------	-------------	--------------------

No Anomaly detectors to display

Identify potential problems early on.

Create Anomaly detector

Learn more

Time

Azure Cognitive Services

Decision –*Anomaly Detector*

Easily embed **time-series anomaly detection capabilities** into your apps to help **users identify problems quickly**.

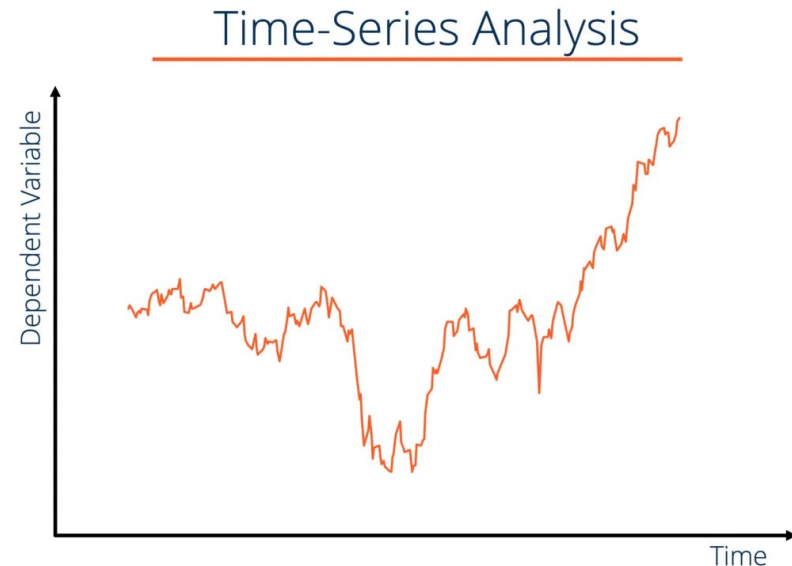
Anomaly Detector **ingests time-series data** of all types and selects the best anomaly detection algorithm for your data to ensure high accuracy.

Detect spikes, dips, deviations from cyclic patterns, and trend changes through both univariate and multivariate APIs.

Time series data

Data have been collected over a period of time on one or more variables.

A Time series is very frequently plotted via a run chart (which is a temporal line chart).



Azure Cognitive Services

Decision –*Anomaly Detector*

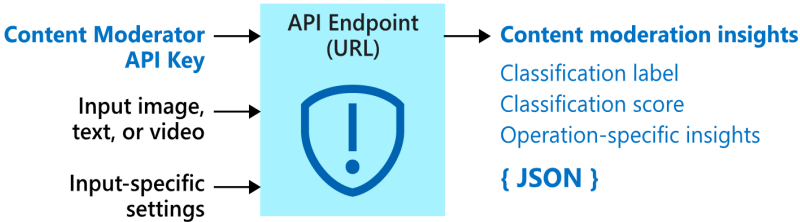
The Anomaly Detector API enables you to monitor and detect abnormalities in your time series data without having to know machine learning. The Anomaly Detector API's algorithms adapt by automatically identifying and applying the best-fitting models to your data, regardless of industry, scenario, or data volume. Using your time series data, the API determines boundaries for anomaly detection, expected values, and which data points are anomalies.



Azure Cognitive Services

Decision –Content Moderator

The Content Moderator service includes Moderation APIs, which check content for material that is potentially inappropriate or objectionable.



API group	Description
Text moderation	Scans text for offensive content, sexually explicit or suggestive content, profanity, and personal data.
Custom term lists	Scans text against a custom list of terms along with the built-in terms. Use custom lists to block or allow content according to your own content policies.
Image moderation	Scans images for adult or racy content, detects text in images with the Optical Character Recognition (OCR) capability, and detects faces.
Custom image lists	Scans images against a custom list of images. Use custom image lists to filter out instances of commonly recurring content that you don't want to classify again.
Video moderation	Scans videos for adult or racy content and returns time markers for said content.

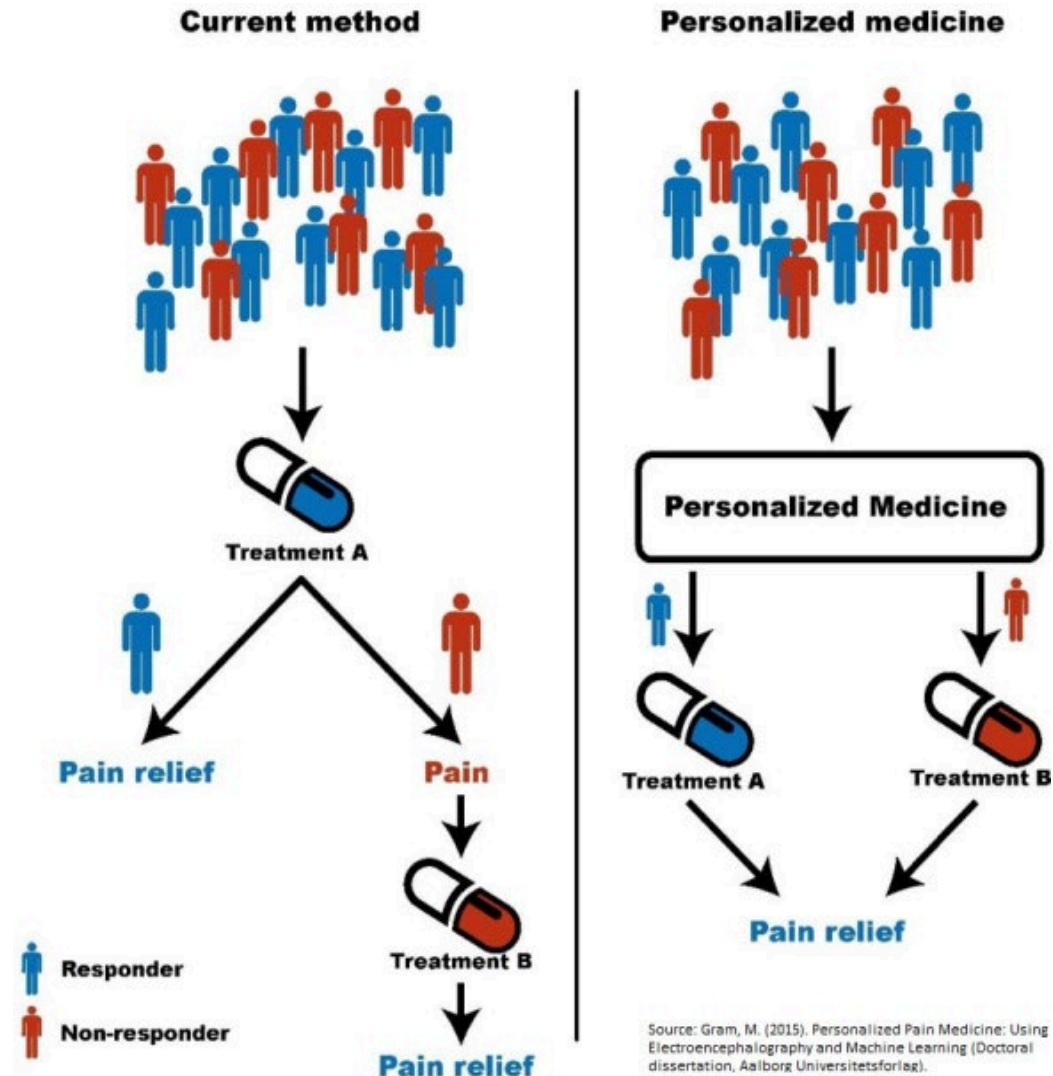
Azure Cognitive Services

Decision –*Personalizer*

Boost usability and user satisfaction with **reinforcement learning–based capabilities**.

Prioritize relevant content, layouts, and conversations through an easy-to-use API.

Use Personalizer as a standalone personalization solution or to complement an existing solution.

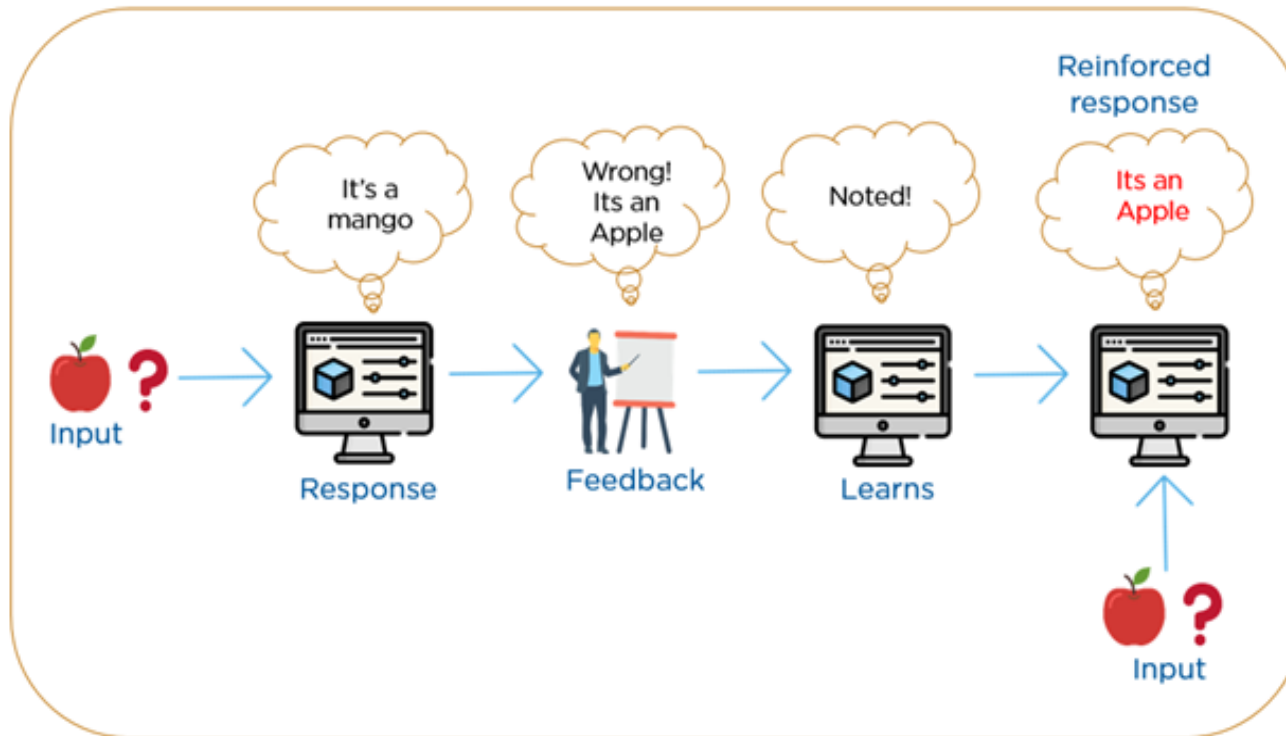


Source: Gram, M. (2015). Personalized Pain Medicine: Using Electroencephalography and Machine Learning (Doctoral dissertation, Aalborg Universitetsforlag).

Azure Cognitive Services

Decision –*Personalizer*

- Personalizer uses **reinforcement learning** to select the best item (action) based on collective behavior and reward scores across all users. Actions are the content items, such as news articles, specific movies, or products.
- The action shown to the user is chosen with machine learning models, that **try to maximize the total amount of rewards over time**.



Language Azure Cognitive Services

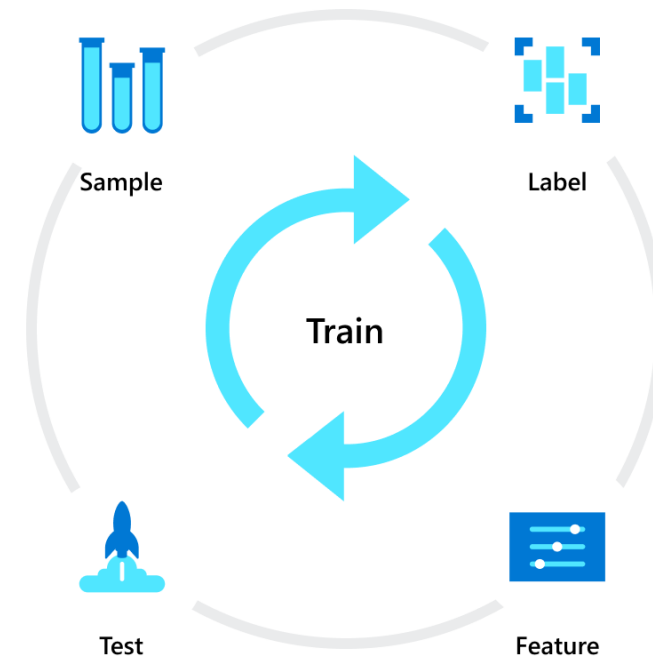


Azure Cognitive Services

Language –*Language Understanding*

- Build applications that can **understand natural language**.
- Using machine learning technology and our visual user interface, developers and subject matter experts can build custom machine-learned language models that interpret user goals and **extract key information from conversational phrases**.

Azure provides prebuilt entities, features, and applications to jump-start your project.



Azure Cognitive Services

Language –*Language Understanding*

- Teach language models like you would teach a person using machine teaching techniques.

These are different ways a person would book a meeting.

"Book a call for Tuesday with Marc..."

"Schedule a 1-1 with Sara tomorrow"



Meetings often have attendees and a date and time

"Schedule an appointment with Sam today at 1pm"



Azure Cognitive Services

Language –*Language Understanding*

- Controlling IoT devices using a voice assistant

Create seamless conversational interfaces that understand natural language with all your internet-accessible devices—from your connected television or fridge to devices in a connected power plant.

turn the right light on

switch floor lamp to green

switch all lights to green

turn the table light off

turn on the left light

all lights off

all on

Smart light application in action



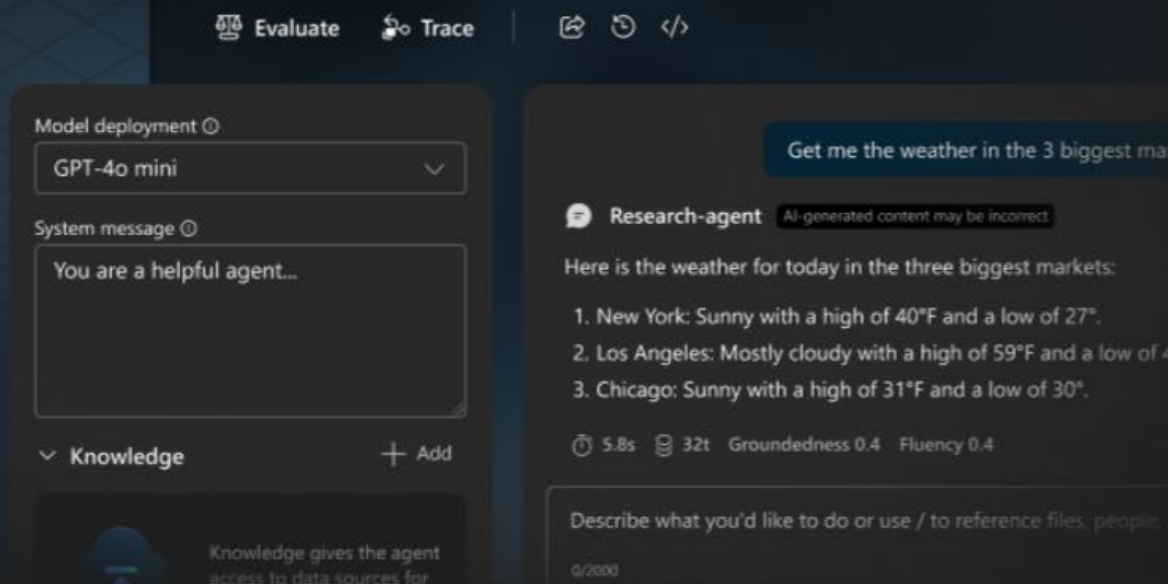
LUIS application response ⓘ

```
{
  "query": "all lights off",
  "topScoringIntent": {
    "intent": "TurnAllOff",
    "score": 0.9947984
  },
  "entities": []
}
```

Create smarter agents with Azure AI Foundry

Design, customize, and manage powerful, adaptable AI agents that automate tasks, integrate seamlessly with your apps, and enhance user experiences.

Create an agent



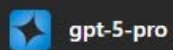
Explore models and capabilities

Choose the right model for your use case

Search for a model

Latest models

[Go to full model catalog](#)



gpt-5-pro

Chat completion



grok-4

Chat completion



grok-4-fast-reasoning

Chat completion



grok-4-fast-non-reasoning

Chat completion



DeepSeek-V3.1

Chat completion



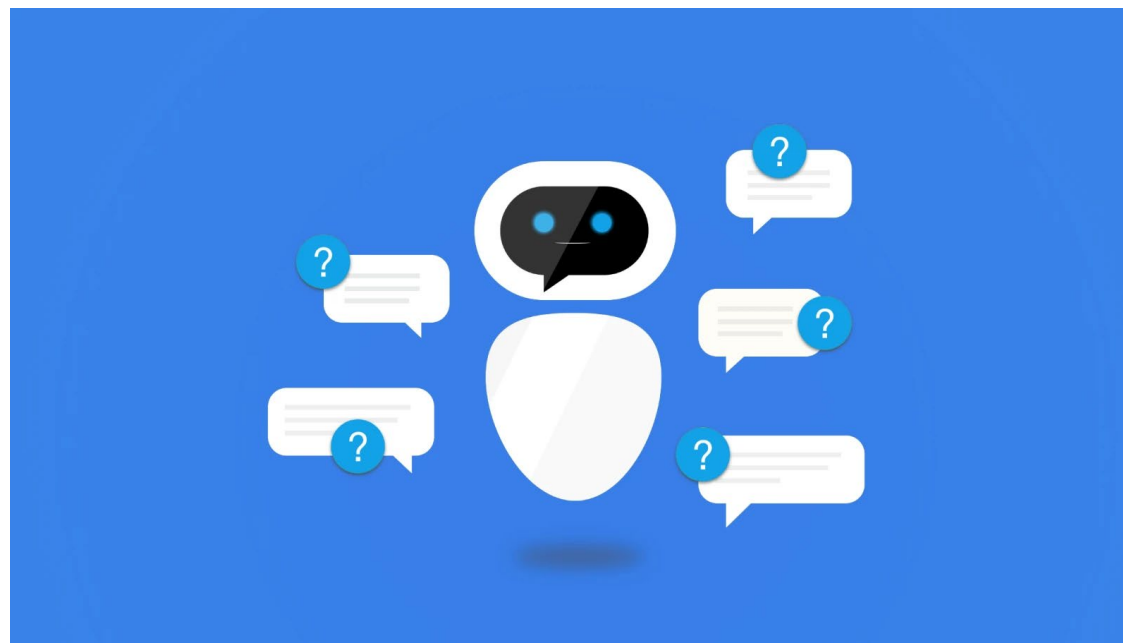
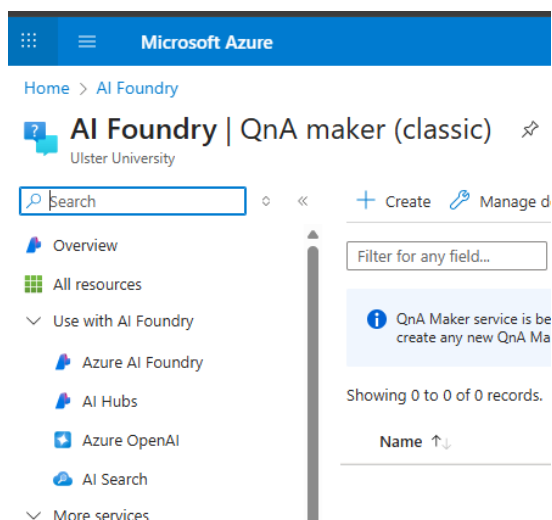
grok-code-fast-1

Chat completion

Azure Cognitive Services

Language –QnA Maker

- QnA Maker is a cloud-based API service that **lets you create a conversational question-and-answer** layer over your existing data.
- Use it to **build a knowledge base by extracting questions and answers from your semi-structured content**, including FAQ, manuals, and documents.
- **Answer users' questions with the best answers from the QnAs** in your knowledge base—automatically.



Azure Cognitive Services

Language –QnA Maker

What is a knowledge base?

QnA Maker imports your content into a knowledge base of question and answer pairs. The import process extracts information about the relationship between the parts of your structured and semi-structured content to imply relationships between the question and answer pairs. You can edit these question and answer pairs or add new pairs.

After you publish your knowledge base, a client application sends a user's question to your endpoint. Your QnA Maker service processes the question and responds with the best answer.

The content of the question and answer pair includes:

- All the alternate forms of the question
- Metadata tags used to filter answer choices during the search
- Follow-up prompts to continue the search refinement

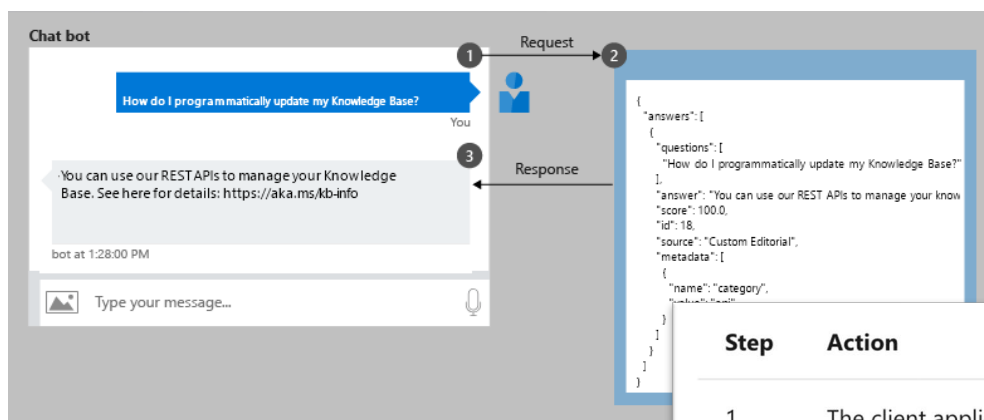
Question	Answer	Metadata tags ?
Original source: https://docs.microsoft.com/en-us/azure/cognitive-services/qnamaker/faqs		
I accidentally deleted a part of my QnA Maker, what should I do? ✕	All deletes are permanent, including question and answer pairs, files, URLs, custom questions and answers, knowledge bases, or Azure resources. Make sure you export your knowledge base from the **Settings** page before deleting any part of your knowledge base.	Type : troubleshooting ✕
Can I undo deleted questions and answers? ✓ ✕		Format : text-only ✕
		Nextstep : recover ✕ +

Azure Cognitive Services

Language –QnA Maker

Create a chat bot programmatically

- Once a QnA Maker knowledge base is published, a client application sends a question to your knowledge base endpoint and receives the results as a JSON response. A common client application for QnA Maker is a chat bot.



Step	Action
1	The client application sends the user's <i>question</i> (text in their own words), "How do I programmatically update my Knowledge Base?" to your knowledge base endpoint.
2	QnA Maker uses the trained knowledge base to provide the correct answer and any follow-up prompts that can be used to refine the search for the best answer. QnA Maker returns a JSON-formatted response.
3	The client application uses the JSON response to make decisions about how to continue the conversation. These decisions can include showing the top answer and presenting more choices to refine the search for the best answer.

Azure Cognitive Services

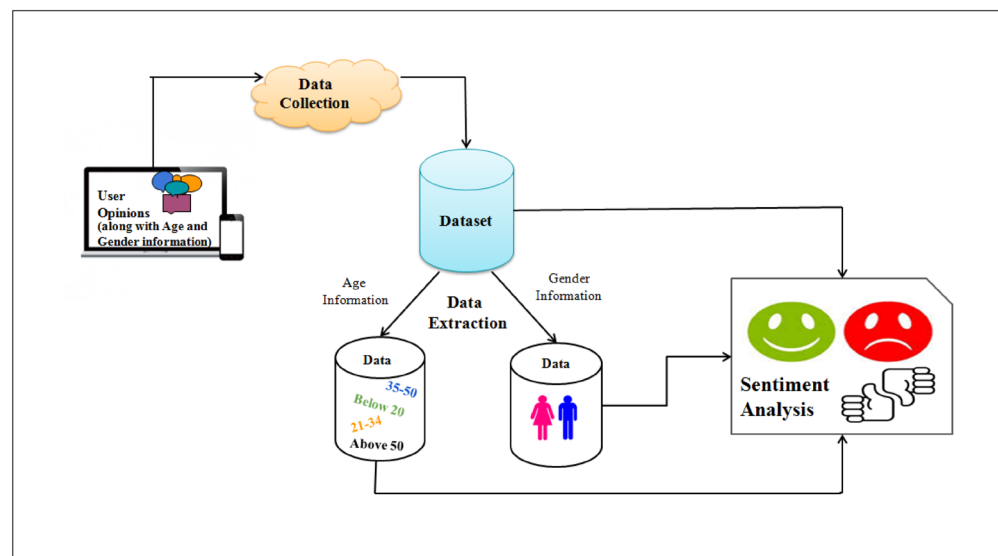
Language – *Text Analytics*

- The Text Analytics API is a cloud-based service that provides **Natural Language Processing (NLP)** features for text mining and text analysis, including:
 - Sentiment analysis
 - Key phrase extraction
 - Language detection
 - Text summarization

Azure Cognitive Services

Language – *Text Analytics*

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 - Sentiment analysis
 - Key phrase extraction
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 - Text summarization



Use sentiment analysis (SA) to process text and classify positive or negative sentiment.

The feature provides sentiment labels (such as "negative", "neutral" and "positive") based on the highest confidence score found by the service at a sentence and document level.

Azure Cognitive Services

Language – *Text Analytics*

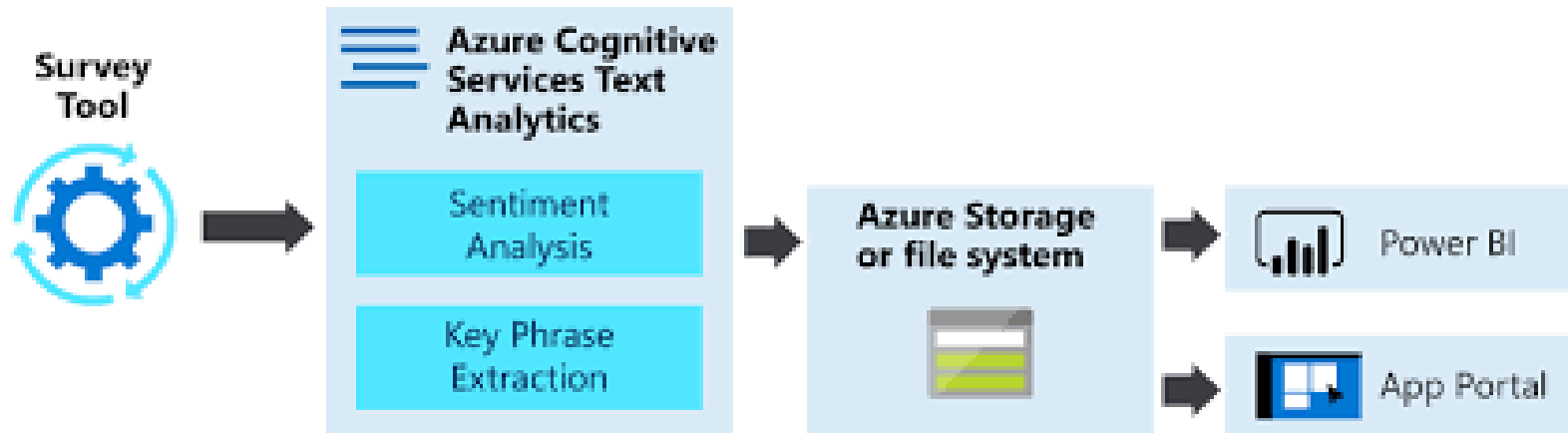
- The Text Analytics API is a cloud-based service that provides **Natural Language Processing (NLP)** features for text mining and text analysis, including:

- Sentiment analysis
- Key phrase extraction
- Language detection
- Text summarization

Use key phrase extraction (KPE) to quickly identify the main concepts in text.

For example, in the text "**The food was delicious and there were wonderful staff**",

Key Phrase Extraction will return the main talking points: "**food**" and "**wonderful staff**".



Azure Cognitive Services

Language – *Text Analytics*

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 - Sentiment analysis
 - Key phrase extraction
 - Language detection
 - Text summarization

```
JSON Copy  
{  
  "documents": [  
    {  
      "id": "1",  
      "text": "This document is in English."  
    },  
    {  
      "id": "2",  
      "text": "Este documento está en inglés."  
    },  
    {  
      "id": "3",  
      "text": "Ce document est en anglais."  
    },  
    {  
      "id": "4",  
      "text": "本文件为英文"  
    },  
    {  
      "id": "5",  
      "text": "Этот документ на английском языке."  
    }  
  ]  
}
```

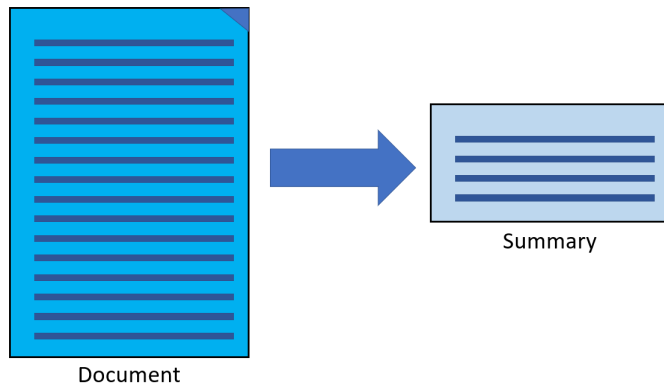
Language detection can detect the language an input text is written in and report a single language code for every document submitted on the request in a wide range of languages, variants, dialects, and some regional/cultural languages.

The language code is paired with a confidence score.

Azure Cognitive Services

Language – *Text Analytics*

- The Text Analytics API is a cloud-based service that provides **Natural Language Processing (NLP)** features for text mining and text analysis, including:
 - Sentiment analysis
 - Key phrase extraction
 - Language detection
 - Text summarization



Summarization produces a summary of text by extracting sentences that collectively represent the most important or relevant information within the original content.

This feature condenses articles, papers, or documents down to key sentences.

Azure Cognitive Services

Language – *Translator*

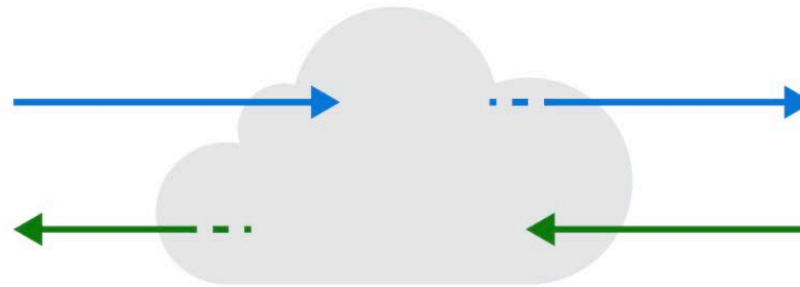
Translate text instantly or in batches across more than 90 languages, powered by the latest innovations in machine translation.

Support a wide range of use cases, such as translation for call centres, multilingual conversational agents, or in-app communication.

Client app or web page

Cet outil de traduction est
un moyen facile d'étendre
la portée de vos contenus
internes et externes

這個翻譯工具是內容的
可以輕鬆地聽到你內部
和外部範圍



Web API





Speech

Azure Cognitive Services

Azure Cognitive Services

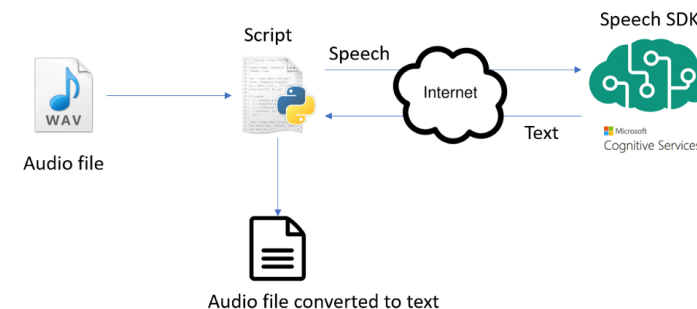
Speech –*Speech to Text*

Quickly and accurately transcribe audio to text in more than 85 languages and variants.

Customize models to enhance accuracy for domain-specific terminology.

The speech-to-text service defaults to using the Universal language model. This model was trained using Microsoft-owned data and is deployed in the cloud. It's optimal for conversational and dictation scenarios

Azure AI – Speech to Text Demo



Azure Cognitive Services

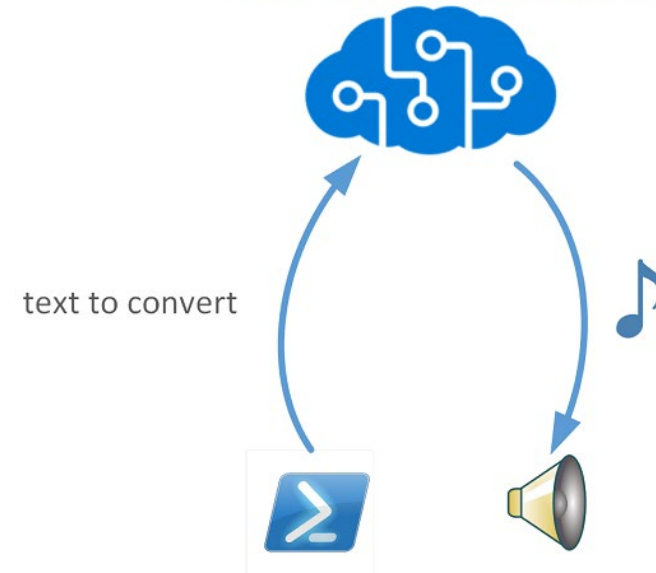
Speech – *Text to Speech*

Build apps and services that speak naturally, choosing from more than 250 voices and over 70 languages and variants.

Differentiate your brand with a customized voice, and access voices with different speaking styles and emotional tones to fit your use case—from text readers to customer support chatbots.

Azure Cognitive Services Text to Speech

Cognitive Text to Speech (TTS)



Azure Cognitive Services

Speech –*Speech Translation*

Translate audio from more than 30 languages and customize your translations for your organization's specific terms—all in your preferred programming language.

Add high-quality translations to your apps

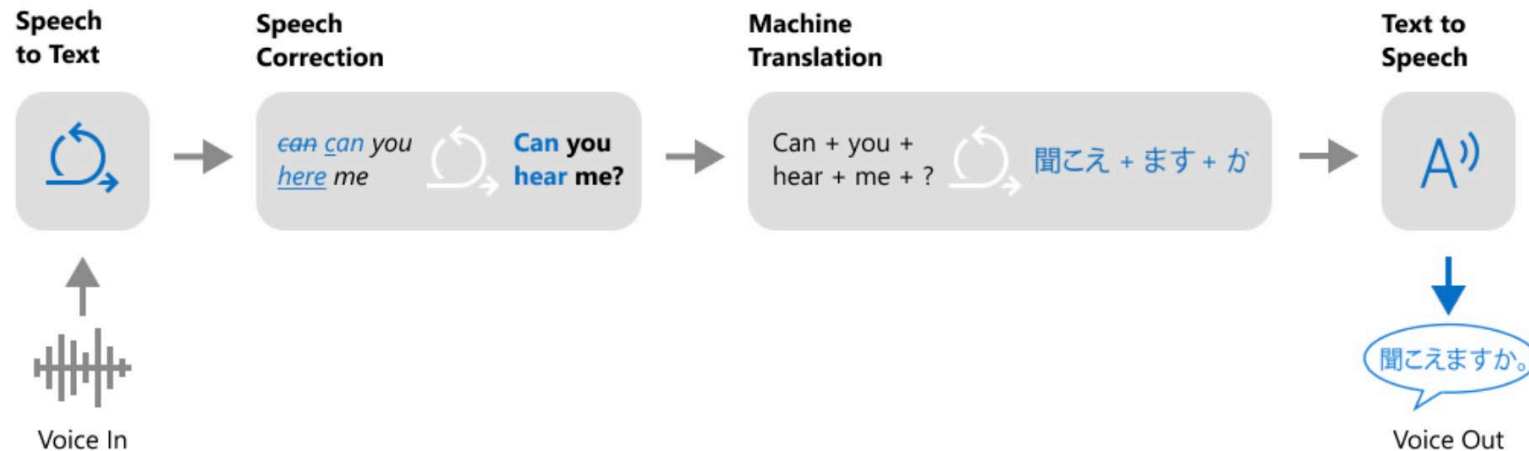
Generate speech-to-speech and speech-to-text translations with a single API call. Speech Translation captures the context of full sentences to provide accurate, fluent translations and improve communication between speakers of different languages.

Tailor translations to reflect domain-specific terminology

[Customize](#) speech recognition and translation for terminology specific to your business or industry. Train and deploy a custom translation system—without requiring machine learning expertise.

Normalize text for better translations

Speech Translation can remove verbal fillers ("um," "uh," and coughs) and repeated words, add proper punctuation and capitalization, and exclude profanities for more readable translations.

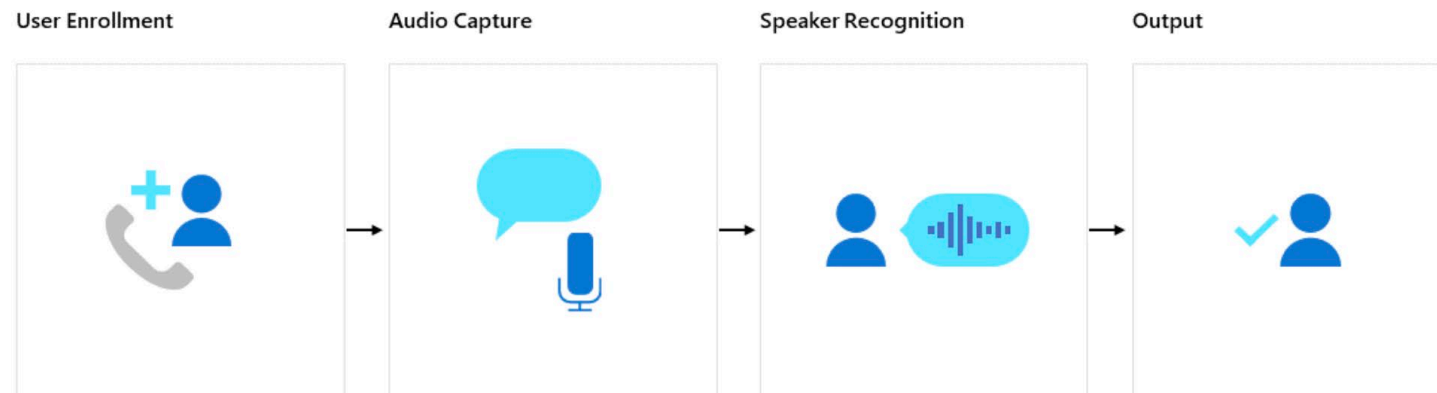


Azure Cognitive Services

Speech –*Speaker Recognition*

Accurately verify and identify speakers by their unique voice characteristics.

The Speaker Recognition service provides algorithms that verify and identify speakers by their unique voice characteristics using voice biometry. Speaker Recognition is used to answer the question “who is speaking?”. You provide audio training data for a single speaker, which creates an enrolment profile based on the unique characteristics of the speaker's voice. You can then cross-check audio voice samples against this profile to verify that the speaker is the same person (speaker verification), or cross-check audio voice samples against a group of enrolled speaker profiles, to see if it matches any profile in the group (speaker identification).

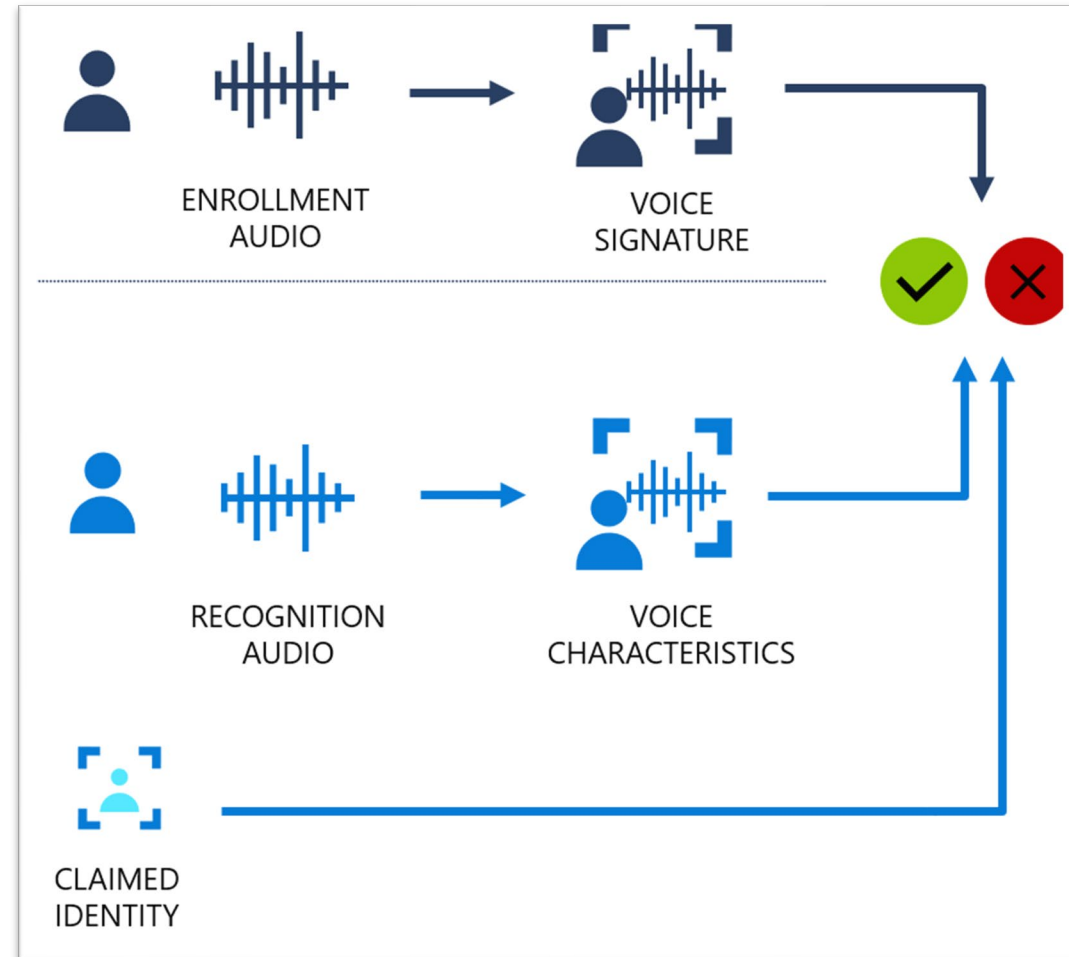


Azure Cognitive Services

Speech –*Speaker Recognition*

Speaker Verification streamlines the process of verifying an enrolled speaker identity with either passphrases or free-form voice input.

It can be used to verify individuals for secure, frictionless customer engagements in a wide range of solutions, from customer identity verification in call centers to contact-less facility access.



Resources

[azure-docs/articles at main · MicrosoftDocs/azure-docs \(github.com\)](#)

github.com/MicrosoftDocs/azure-docs/tree/master/articles/cognitive-services

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<> Code

Issues3.8k

Pull requests306

Security

Insights

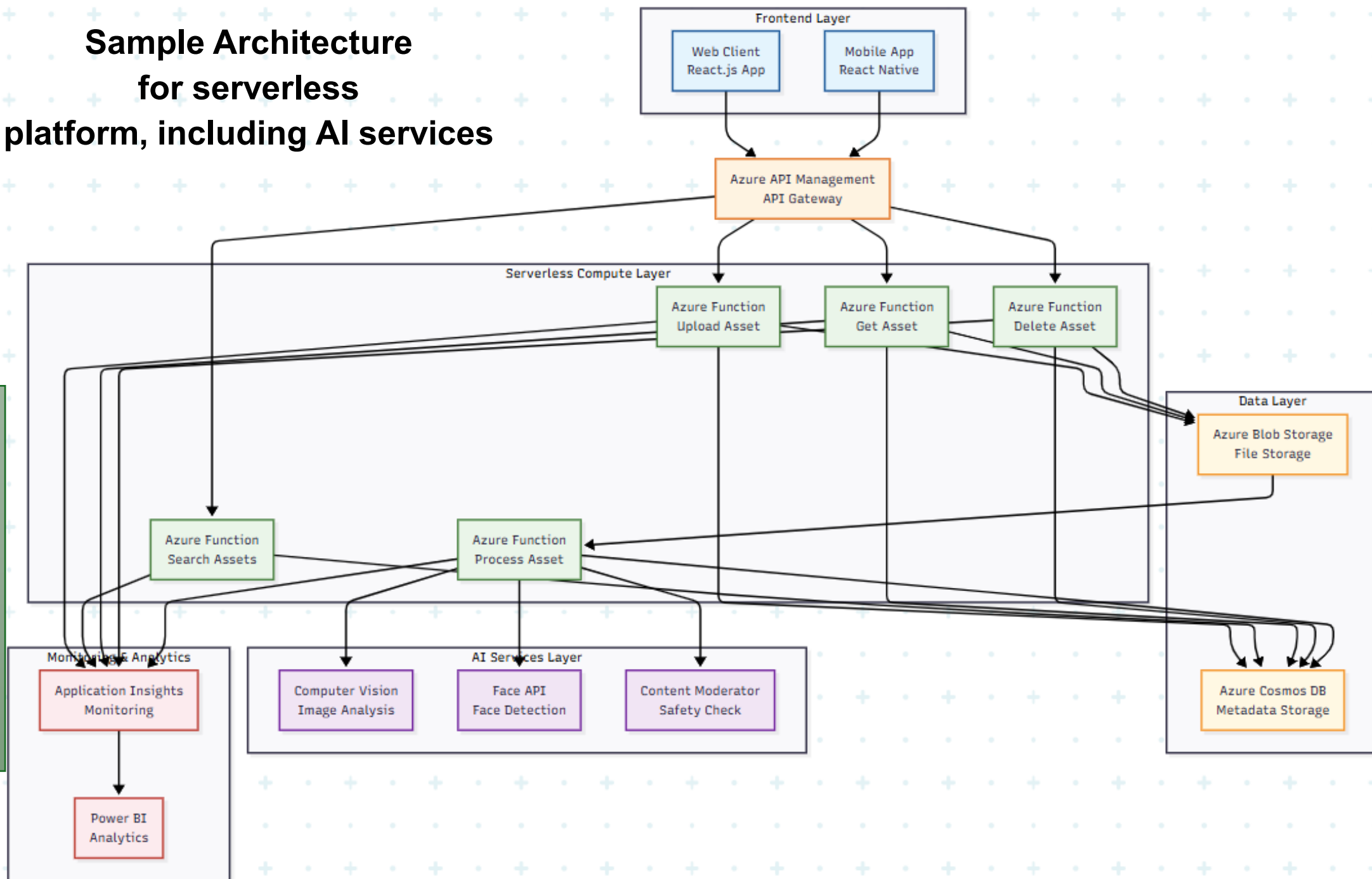
masterazure-docs / articles / cognitive-services /Go to file

PRMerger8 Merge pull request #174946 from jfilcik/patch-204dfded 22 hours agoHistory

..

📁 Anomaly-Detector	Fixing preserve-view-not-set	14 days ago
📁 Bing-Autosuggest	Resolve live publish PR issue	23 days ago
📁 Bing-Custom-Search	Resolve live publish PR issue	23 days ago
📁 Bing-Entities-Search	Fix absolute links	14 days ago
📁 Bing-Image-Search	Resolve live publish PR issue	23 days ago
📁 Bing-News-Search	Resolve live publish PR issue	23 days ago
📁 Bing-Spell-Check	Resolve live publish PR issue	23 days ago
📁 Bing-Video-Search	Resolve live publish PR issue	23 days ago
📁 Bing-Web-Search	Resolve live publish PR issue	23 days ago
📁 Computer-vision	Merge pull request #174946 from jfilcik/patch-2	22 hours ago
📁 Content-Moderator	Merge pull request #173875 from PatrickFarley/freshness-pass	13 days ago
📁 Custom-Vision-Service	Merge pull request #173875 from PatrickFarley/freshness-pass	13 days ago
📁 Encryption	fix broken relative links	2 months ago
📁 Face	Merge pull request #172461 from PatrickFarley/face-updates	2 days ago
📁 LUIS	Merge pull request #174210 from PatrickFarley/minor-updates	13 days ago
📁 QnAMaker	Fix links	14 days ago
📁 Speaker-recognition	Links: Azure - cognitive-services\speaker-recognition	12 months ago
📁 Speech-Service	Merge pull request #82063 from jimxie.Jjimxie/update_diarization_feature	5 days ago
📁 Translator	Edit spelling	2 days ago
📁 Video/Images	Delete HowtoCallVideoAPIs.md	4 years ago
📁 big-data	Update getting-started.md	3 months ago
📁 bing-local-business-search	DocuTune: add ms.custom tags and fix formatting	7 months ago
📁 bing-visual-search	Resolve live publish PR issue	23 days ago
📁 containers	Update container-image-tags.md	12 days ago
📁 includes	broken link fix	2 months ago
📁 media	More updates	7 months ago
📁 personalizer	Update concepts-feature.md	15 days ago

Sample Architecture for serverless platform, including AI services



An inspiration for
your CW1
architecture, if you
are choosing
serverless.

Note: You don't
need to add all of
these components.
Only add those
components which
are related to your
use case.



Questions

